NINGARAJ R UKKALI

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Summary

BCA graduate and aspiring Software Engineer with hands-on experience building backend and full-stack applications involving REST APIs, real-time communication, databases, and containerized services. Strong foundation in Python, Java, JavaScript, SQL, OOP, and software development fundamentals. Experienced with FastAPI, Flask, WebSockets, Redis Pub/Sub, PostgreSQL, React.js, Docker, and Linux development environments. Interested in building reliable software systems, solving performance and scalability problems, and learning functional programming and large-scale system design.

Education

Bachelor of Computer Applications (BCA) 2023 – 2026
Mangalore University, Mangalore, Karnataka, India CGPA: 8.1

Technical Skills

Programming	: Python, Java, JavaScript (ES6+), SQL
Software Engineering	: OOP, Data Structures & Algorithms, REST APIs, API Integration, Authentication, Problem Solving
Backend & Systems	: FastAPI, Flask, WebSockets, Redis Pub/Sub, Backend Services
Databases	: PostgreSQL, MySQL, MongoDB, Redis
Frontend	: React.js, Tailwind CSS, Responsive UI, Component-Based Development
Infrastructure & Tools	: Docker, Linux CLI, Git, GitHub, Postman, VS Code

Projects

PulseSim – Real-Time Medical Simulation Platform [GitHub]

Technologies: Python, FastAPI, WebSockets, Redis, PostgreSQL, Docker, LangGraph, React.js

- Developed a backend-driven simulation platform supporting real-time patient state updates and interactive user workflows.
- Implemented bidirectional real-time communication using FastAPI, WebSockets, and Redis Pub/Sub.
- Designed state evaluation workflows to manage changing patient conditions and simulation events.
- Persisted chronological session events in PostgreSQL for session analysis and evaluation.
- Containerized backend services with Docker and developed using Linux-based tooling.

CraftCycle | AI-Powered Circular Economy Marketplace [GitHub] | [Live Demo]

Technologies: HTML, CSS, JavaScript, Python, Flask, PostgreSQL, Supabase, REST APIs, JWT

- Developed a full-stack marketplace for sourcing scrap materials and creating/upcycling products.
- Built Flask REST APIs for authentication, marketplace workflows, and application data management.
- Integrated Google Gemini Vision for AI-assisted image analysis and upcycling recommendations.
- Connected the frontend with backend APIs and PostgreSQL/Supabase services.

Work Experience

Freelance Software Developer | Self-Employed 2025 – 2026
Mangalore, Karnataka

- Developed full-stack applications for student clients based on functional requirements and project specifications.
- Worked across requirements analysis, application development, API integration, database integration, debugging, testing, and deployment.
- Developed backend services and REST APIs using Python, FastAPI, and Flask.
- Built responsive interfaces using React.js, JavaScript, and Tailwind CSS.
- Used PostgreSQL, MySQL, MongoDB, Redis, Git, Docker, and Linux-based development workflows.

Activities & Achievements

- Participated in a Hackathon, developing an end-to-end functional MVP under strict time constraints
- Participated in various University Events, technical symposiums, and coding competitions
- Demonstrated teamwork, problem-solving, and communication skills across technical and client projects
- Maintained strong academic performance with an 8.1 CGPA in Bachelor of Computer Applications

Certifications

- Generative AI Professional Certification — Issued by OutSkill AI
- Foundations: Data, Data, Everywhere — Issued by Google (Coursera)